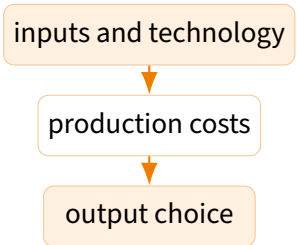
Unit 3: Production and Costs

How firms choose inputs and output

- production, productivity, and scale;
- the cheapest way to produce;
- short-run and long-run costs;
- profit maximization and firm supply.



The Other Side of the Market

In Units 1 and 2, we studied how consumers make choices. We now turn to the decisions of firms.

Consumers

Preferences, income, and prices determine what consumers choose to buy.

Consumer choices $\longrightarrow$ Demand

Firms

Technology, costs, and prices determine what firms choose to produce and sell.

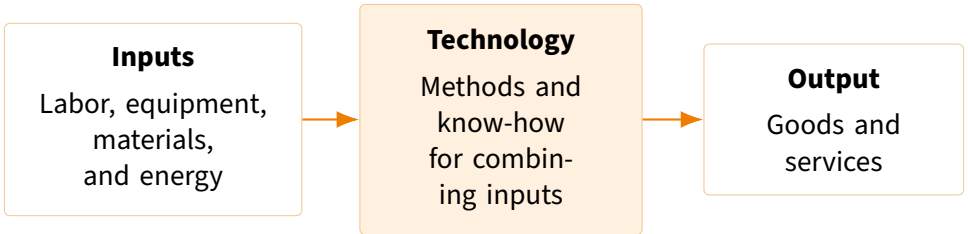
Firm choices $\longrightarrow$ Supply

Bringing these decisions together will help us understand **how markets allocate goods and determine prices**.

In this unit, we build the supply side. In Unit 4, we bring supply and demand together to study market equilibrium.

Inside the Firm: Turning Inputs into Output

To understand supply, we need to look inside the firm: how does it produce the goods and services it sells?



- Different input combinations can produce the same output.
- The available technology determines what is feasible.
- We summarize these possibilities with a **production function**.

The Production Function

$$\boxed{q = f(L, K; A)}$$

- q: output per period, at a specified quality.
- L: labor services, such as worker-hours.
- K: capital services, such as machine-hours.
- A: technology and productive know-how.

The production function describes the **maximum feasible output** from a given input bundle, holding technology fixed.

We begin with two inputs to keep the analysis simple. The same framework can include materials, fuel, **Data** *(Isaac's part)*, or other inputs.

An observed firm may produce below this frontier because of downtime, waste, or poor organization.

Returns to Scale

What happens if the firm scales up **all inputs together**?

$$(L, K) \longmapsto (tL, tK), \qquad t > 1.$$

Returns to scale	Output comparison	Doubling all inputs
Increasing	$f(tL, tK) > tf(L, K)$	More than doubles output
Constant	$f(tL, tK) = tf(L, K)$	Doubles output
Decreasing	$f(tL, tK) < tf(L, K)$	Less than doubles output

This is a **technology comparison**: input proportions and productive know-how stay fixed.

One Input versus All Inputs

Consider $q = \sqrt{LK}$.

Increase labor alone:

$$f(2L, K) = \sqrt{2}\, f(L, K).$$

Output rises less than proportionally as labor crowds the fixed capital.

Double both inputs:

$$f(2L, 2K) = 2f(L, K).$$

The whole operation can be replicated.

L	K	$q = \sqrt{LK}$
4	4	4
8	4	$4\sqrt{2} \simeq 5.66$
8	8	8

Different experiments

Diminishing marginal product changes one input. Returns to scale change all inputs.

Firms' Decisions

We study firms that take market prices as given:

w : labor price, r : capital rental price, p : output price.

How should the firm produce?

Choose the input combination that produces a given quantity at the lowest cost.

How much should it produce?

Choose the quantity that gives the highest profit at the selling price.

$$\underbrace{\pi}_{\text{Profit}} = \underbrace{pq}_{\text{Revenue}} - \underbrace{(wL + rK)}_{\text{Input costs}}, \qquad q = f(L, K; A).$$

Technology and prices jointly determine the firm's profit-maximizing choices.

We will use this framework to make sense of data on production, costs, and profitability.

Data Notebook: Why Do Plants Close?

Sections 0–3 | Explore the evidence

A claim to evaluate:

> "The new emissions charge is forcing power plants to close. Removing it will keep them in business."

- Open `unit3_power_plants_practical.ipynb`.
- Work through **Sections 0–3**, stopping at **STOP FOR THEORY**.
- Describe what changed before deciding why it changed.

All data and policy details are synthetic. The question is inspired by Linn and McCormack (2019), *The RAND Journal of Economics*.

What Does One Observation Mean?

Sections 0–1 | Data structure

60 plants $\times$ 730 days = 43,800 plant-day records.

File	One row	What we observe
`plant_daily.csv`	Plant-day	Output, fuel, emissions, bills
`market_days.csv`	Day	Electricity price, fuel price, charge
`plants.csv`	Plant	Capacity, cost rates, commitments
`renewal_decisions.csv`	Plant	Renewal for 2023

Merging creates `d`. Idle days stay in the sample with zero output.

All plants have committed for 2021 and 2022. The separate renewal decision concerns **2023**; 30 plants do not renew.

From Measurements to Plant Accounts

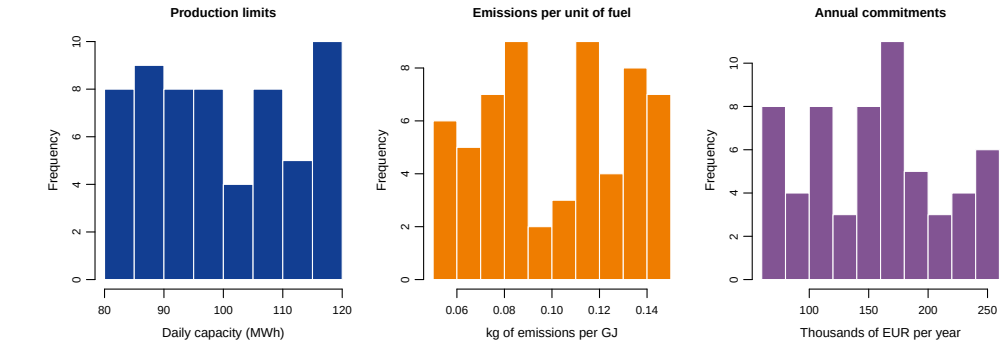
Output is in **MWh**; fuel in **GJ**; emissions in **kg**.

Measure	Construct it from the data
Capacity use	Daily output / daily capacity
Revenue	Electricity price $\times$ output
Fuel bill	Fuel price $\times$ fuel
Emissions bill	Charge per kg $\times$ emissions
Variable cost	Fuel bill + emissions bill + other variable bill
Operating contribution	Revenue $-$ variable cost

The annual commitment is a separate bill. Historical construction spending was paid in the past. We will ask which costs matter for each decision.

Meet the Plants

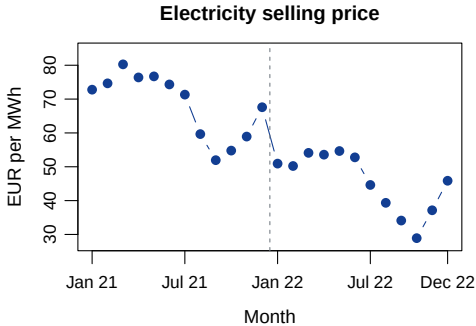
Section 1 | Plant characteristics

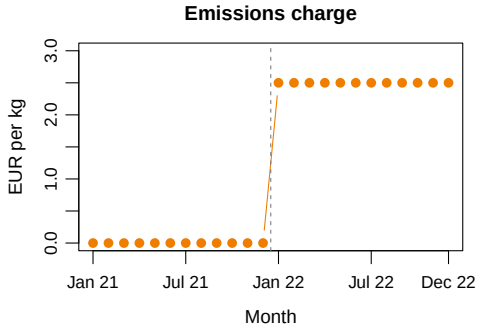


The plants differ in capacity, emissions intensity and annual commitments. Each plant appears once in these distributions.

Market Conditions Over Our Sample

Section 2 | Monthly market conditions

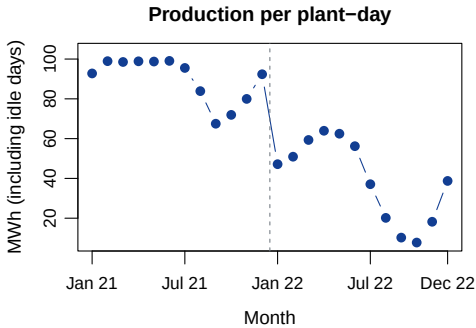


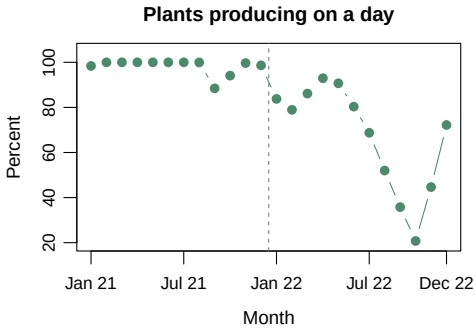


January 2022 brings a new emissions charge. Prices have been declining. A before–after comparison combines their effects.

Production and Operating Days Fall

Section 2 | Monthly activity

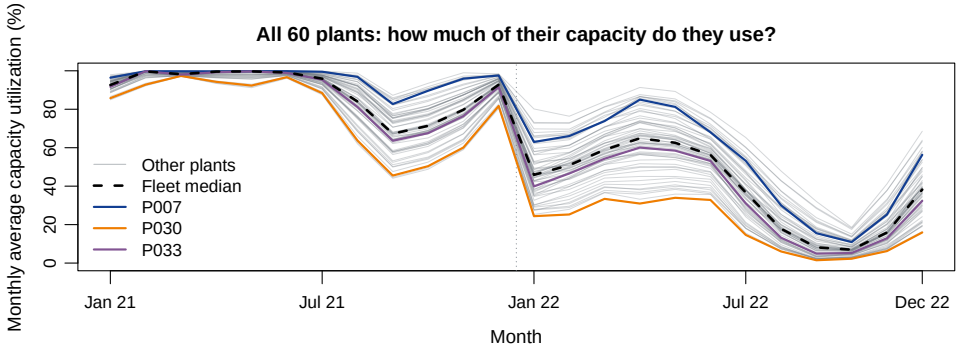




Plants produce less, and more plant-days have zero output. The sample still contains the same 60 plants.

Follow All 60 Plants

Section 2 | Monthly capacity use



Thin lines track individual plants; the dashed line is the fleet median. Lower utilization means less use of existing capacity.

Does the Year's Production Cover the Annual Bill?

For one plant and one year, add up the daily accounts:

$$Q = \sum_t q_t, \qquad V = \sum_t VC_t, \qquad R = \sum_t p_t q_t.$$

Then calculate

$$\text{profit} = R - V - F, \qquad \text{average total cost} = \frac{V + F}{Q}.$$

$$\text{fuel per MWh} = \frac{\sum_t B_t}{Q}.$$

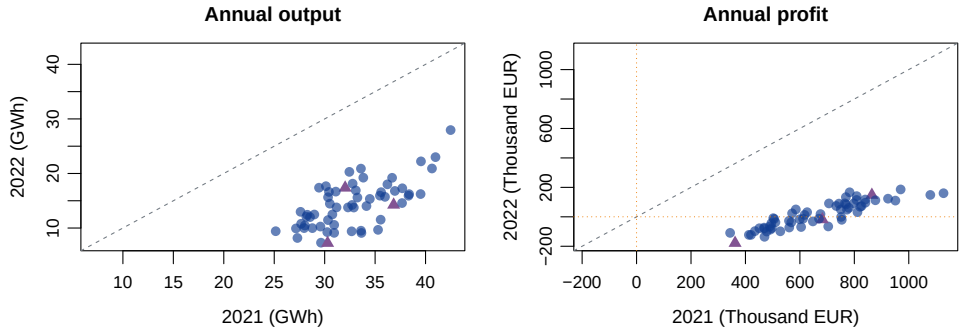
Count the annual commitment F **once**. Daily data describe operation; annual accounts compare the year's earnings with the year's bill.

Puzzle Across All 60 Firms

Observed outcome	2021	2022
Output (GWh)	1,967	860
Mean operating days	359	245
Plants with annual losses	0	30
Total annual profit (EUR million)	40.04	0.83
Average total cost (EUR/MWh)	50.50	56.74
Fuel use (GJ/MWh)	14.16	13.15

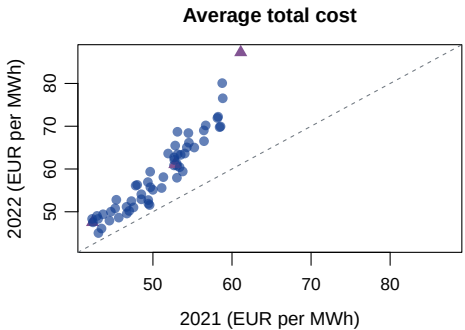
Output and profits fall. Fuel used per MWh also falls, yet average total cost rises. What could explain all three patterns?

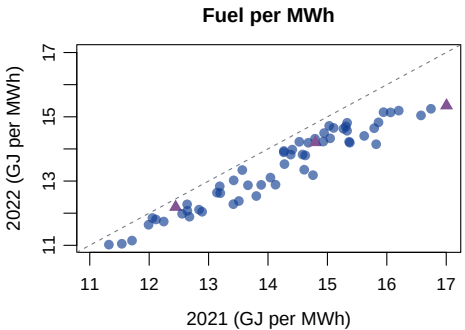
Is the Decline Shared Across Plants?



Each dot compares one plant with itself. Below the diagonal means a lower outcome in 2022; the changes differ across plants.

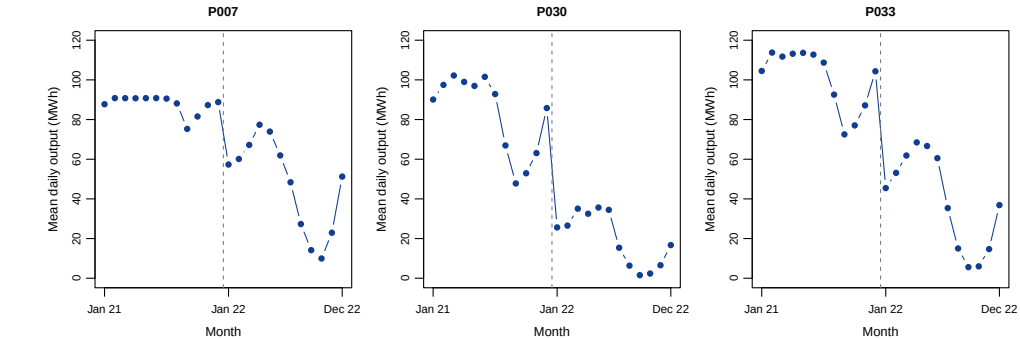
Less Fuel per MWh, but Higher Cost





Every plant uses less fuel per MWh and has higher average total cost. This is a within-plant puzzle, not just a change in the fleet mix.

Three Histories to Keep in Mind



P007, P030 and P033 all reduce production. We will use the same three plants to understand daily output and renewal.

Why Produce at an Annual Loss?

2022 account	P007	P030	P033
Days producing	266	179	219
Contribution (EUR thousand)	218.9	64.5	146.2
Annual commitment (EUR thousand)	70.9	242.9	164.1
Annual profit (EUR thousand)	148.0	-178.4	-17.9
Renew for 2023?	Yes	No	No

P030 and P033 produce on many days, report an annual loss, then decline to renew.

Why can producing today and closing next year both make sense?

Back to Theory: What Must We Explain?

Stop after Section 3

Pause the notebook at STOP FOR THEORY. Write a short initial memo.

1. Why can fuel per MWh fall without a better technology?
2. Why can average total cost rise when output falls?
3. When should a plant produce, idle, or decline to renew?
4. How can we separate the effects of prices and policy?

We need a model connecting **technology** $\rightarrow$ **costs** $\rightarrow$ **choices**. Then we can predict what the same plants would do under different conditions.

From Consumer Choice to Input Choice

Unit 1: choose a consumption bundle

$$\max_{x_1, x_2 \geq 0} u(x_1, x_2) \quad \text{s.t. } p_1 x_1 + p_2 x_2 \leq m.$$

Fix expenditure; reach the highest attainable indifference curve.

$$\frac{MU_1}{MU_2} = \frac{p_1}{p_2}.$$

Unit 3: choose an input bundle

$$\min_{L, K \geq 0} wL + rK \quad \text{s.t. } f(L, K) \geq \bar{q}.$$

Fix output; reach the lowest attainable isocost line.

$$\frac{MP_L}{MP_K} = \frac{w}{r}.$$

The same logic: at a smooth interior optimum, the **technical or subjective trade-off** equals the **market trade-off**.

Here both inputs can adjust. The firm's output target will be chosen in a second step.

Two Steps to Maximum Profit

1. **Choose inputs for each possible output:**

$$C(q; w, r) = \min_{L,K \geq 0} \{wL + rK : f(L, K) \geq q\}.$$

   This step uses technology and input prices.

2. **Choose the output that gives the highest profit:**

$$\max_{q \geq 0} \{pq - C(q; w, r)\}.$$

   This step also needs the selling price p.

The first step supplies a **cost schedule**, not the firm's final output. The second step compares the revenue and cost of each quantity.

Same Output, Different Inputs

We now ask a different question:

How can we produce a given quantity using different combinations of labor and capital?

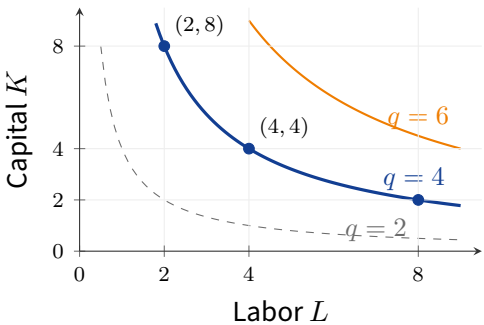
- Fix an output target $\bar{q}$ and technology A.
- Allow **both labor and capital** to adjust.
- Find the input combinations that deliver the target:

$$f(L, K; A) = \bar{q}.$$

These combinations form an **isoquant**.

First identify the feasible input combinations. Then use input prices to find the cheapest one. We will choose the most profitable **output target** afterwards.

Isoquants



An **isoquant** collects input bundles that produce the same output.

$$f(L, K) = \bar{q}.$$

For $q = \sqrt{LK}$, the $q = 4$ isoquant is

$$LK = 16.$$

More labor can compensate for less capital.

Higher isoquants represent more physical output.

The Opportunity Cost of Inputs

Producing uses resources that could have been used elsewhere.

- Hired labor costs the wage w per hour.
- Capital services cost the rental or user cost r per period.
- An owner's time and an owned building also have alternative uses.

$$\text{Input cost} = wL + rK.$$

A resource can have an **economic cost even without a cash payment**. Economic profit subtracts the value of the opportunities the firm gives up.

The user cost r includes the cost of tying up capital and depreciation. It is not simply an interest rate.

Marginal Products and Isoquant Slopes

For a two-input production function, each marginal product changes **one input at a time**:

$$MP_L = \frac{\partial f(L,K)}{\partial L}, \qquad MP_K = \frac{\partial f(L,K)}{\partial K}.$$

For $q = \sqrt{LK}$,

$$MP_L = \frac{1}{2}\sqrt{\frac{K}{L}}, \qquad MP_K = \frac{1}{2}\sqrt{\frac{L}{K}}, \qquad \frac{MP_L}{MP_K} = \frac{K}{L}.$$

Both inputs are **choices**. Holding one fixed when taking a derivative isolates the contribution of the other; it does not impose a short-run constraint.

We combine these two local effects to move along an isoquant.

Technical Substitution

Along an isoquant, output stays constant:

$$dq = \mathrm{MP}_L \, dL + \mathrm{MP}_K \, dK = 0.$$

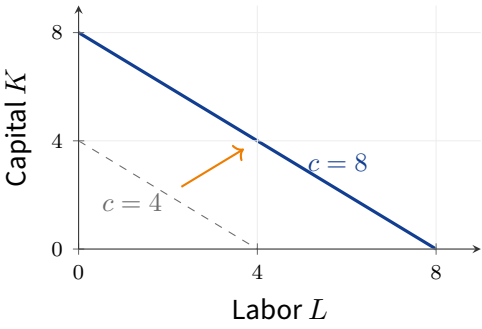
Therefore,

$$\left.\frac{dK}{dL}\right|_{q=\bar{q}} = -\frac{\mathrm{MP}_L}{\mathrm{MP}_K}, \qquad \mathrm{MRTS}_{LK} \equiv \frac{\mathrm{MP}_L}{\mathrm{MP}_K}.$$

MRTS_{LK} measures how much capital the firm can give up for one extra unit of labor, **locally and at unchanged output**.

If $\mathrm{MP}_L = 6$ and $\mathrm{MP}_K = 3$, a small increase in labor can replace about twice as much capital.

Isocost Lines



An **isocost** contains bundles with the same expenditure:

$$wL + rK = c.$$

Equivalently,

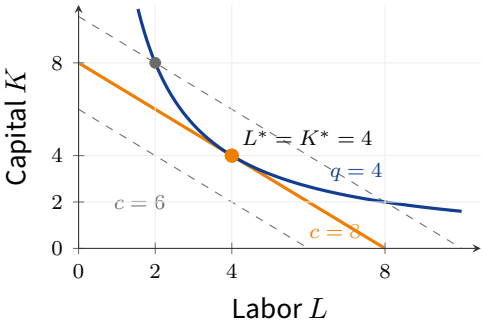
$$K = \frac{c}{r} - \frac{w}{r}L.$$

Its slope is $-w/r$.

At given input prices, lower parallel lines represent lower costs.

In the figure: $w = r = 1$.

The Cheapest Feasible Input Bundle



Target: $q = 4$ with $q = \sqrt{LK}$ and $w = r = 1$.

- Cost 6: no bundle reaches the target.
- Cost 10: $(2, 8)$ works, but is expensive.
- Cost 8: $(4, 4)$ reaches the target at minimum cost.

At a smooth interior optimum,

$$\mathrm{MRTS}_{LK} = \frac{w}{r}.$$

Technology and prices imply the same local trade-off.

Equal Marginal Product per Euro

The tangency condition can also be written as

$$\frac{\mathrm{MP}_L}{w} = \frac{\mathrm{MP}_K}{r}.$$

Each side measures the extra output from one more euro spent on that input.

Suppose instead that

$$\frac{\mathrm{MP}_L}{w} = 3, \qquad \frac{\mathrm{MP}_K}{r} = 1.$$

Moving a small amount of spending from capital to labor raises output at unchanged cost. The original input mix cannot be optimal.

With both inputs in use, a smooth optimum leaves no gain from **reallocating the last euro**.

Cost Minimization: The Lagrangian

For a smooth interior solution with a binding output target,

$$\mathcal{L} = wL + rK + \mu\,[q - f(L, K)].$$

The first-order conditions are

$$w = \mu \mathrm{MP}_L, \qquad r = \mu \mathrm{MP}_K, \qquad f(L, K) = q.$$

Dividing the first two equations gives $\mathrm{MP}_L / \mathrm{MP}_K = w/r$.

The multiplier also has a useful interpretation:

$$\mu = \frac{\partial C(q; w, r)}{\partial q} = \mathrm{MC}.$$

Relaxing the output target by one small unit saves approximately μ euros. Tightening it costs approximately μ euros.

Work Out the Cheapest Input Mix

Target: $\bar{q} = 4$, technology $q = \sqrt{LK}$, input prices $w = 4, r = 1$.

1. Tangency chooses the **proportions**:

$$\frac{K}{L} = \frac{w}{r} = 4 \quad \Longrightarrow \quad K = 4L.$$

2. The output target chooses the **size of the bundle**:

$$LK = 4^2 = 16 \quad \Longrightarrow \quad 4L^2 = 16.$$

3. Hence $L^c = 2$, $K^c = 8$, and minimum cost is

$$C(4) = 4(2) + 1(8) = 16.$$

Tangency alone gives a ratio, not a quantity of output. We used the **given target** to find L and K.

One Example: The Optimal Input Mix

For $q = \sqrt{LK}$, we have $\text{MRTS}_{LK} = K/L$.

$$\frac{K}{L} = \frac{w}{r}, \qquad LK = q^2.$$

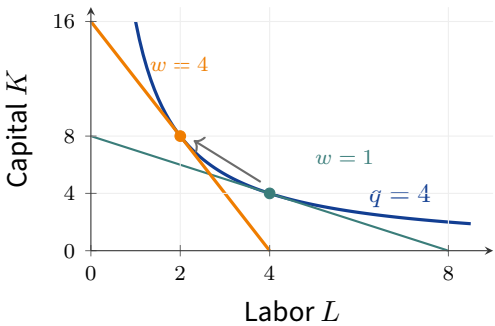
Together these imply the **conditional input demands**

$$L^c(q, w, r) = q\sqrt{\frac{r}{w}}, \qquad K^c(q, w, r) = q\sqrt{\frac{w}{r}}.$$

Target $q = 4$	w	r	L^c	K^c
Equal input prices	1	1	4	4
More expensive labor	4	1	2	8

"Conditional" means that the output target is held fixed.

Substitution after a Wage Increase



The firm stays on the same isoquant but chooses a steeper isocost tangent.

It uses less labor and more capital to produce the same quantity.

Minimum cost rises:

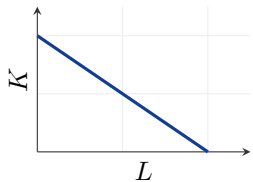
$$4 + 4 = 8 \quad \longrightarrow \quad 4(2) + 8 = 16.$$

Substitution limits the cost increase. Keeping the old mix would cost 20.

Technology Determines Substitutability

Perfect substitutes

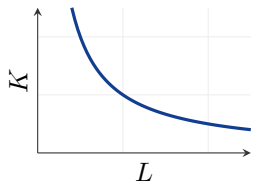
$q = L + K$



One input can replace the other at a constant rate.

Smooth substitution

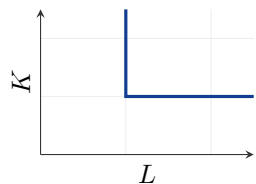
$q = \sqrt{LK}$



Replacing increasingly scarce capital requires more labor.

Fixed proportions

$q = \min\{L, K\}$



Extra labor alone cannot replace missing equipment.

Corners and Fixed Proportions

Perfect substitutes:

$$q = L + K.$$

With $w = 2$ and $r = 1$, capital produces the same output at half the cost.

$$L^c = 0, \qquad K^c = q.$$

The optimum is a **corner**. Equality of marginal products per euro is unnecessary for an unused input.

Fixed proportions:

$$q = \min\{L, K\}.$$

Both inputs are needed together.

$$L^c = K^c = q.$$

The optimum is at the **kink**. A higher wage changes cost but not the input mix at a fixed output target.

The general principle is the cheapest feasible bundle. Tangency describes the smooth interior case.

The Cost Function

Once the firm has found the cheapest inputs,

$$C(q; w, r) = wL^c(q, w, r) + rK^c(q, w, r).$$

For $q = \sqrt{LK}$,

$$C(q; w, r) = 2q\sqrt{wr}.$$

- Output q determines how much must be produced.
- Input prices determine how expensive feasible techniques are.
- Technology determines which techniques are feasible.

A cost function reports the **minimum attainable cost** of each output level. Observed spending may exceed it.

Price Taking and Profit Maximization

Step 2: use the minimum-cost schedule to choose output.

A small firm can sell its output at market price p without affecting that price.

$$R(q) = pq, \qquad \pi(q) = pq - C(q), \qquad MC(q) = C'(q).$$

An extra unit brings in p and costs approximately $\mathrm{MC}(q)$.

- If $p > \mathrm{MC}(q)$, a small increase in output raises profit.
- If $p < \mathrm{MC}(q)$, a small reduction in output raises profit.

At a smooth positive interior optimum, $p = \mathrm{MC}(q)$. We must still check that this choice beats other quantities, including zero.

Constant Costs Do Not Pin Down Scale

Our input-mix example $q = \sqrt{LK}$ implies

$$C(q) = 2q\sqrt{wr}, \qquad MC = 2\sqrt{wr}, \qquad \pi(q) = (p - 2\sqrt{wr})q.$$

With no capacity limit or other costs:

Selling price	Best output choice
$p < 2\sqrt{wr}$	Zero output
$p = 2\sqrt{wr}$	Every output gives zero profit
$p > 2\sqrt{wr}$	No finite optimum in this model

To get a well-defined positive output choice, the next exercise uses a technology with **increasing marginal cost**.

Exercise: The Whole Firm Problem

A price-taking firm has

$$q = (LK)^{1/4}, \qquad w = r = 1, \qquad p = 8.$$

Both inputs can adjust; there are no additional costs or capacity constraints.

1. For a given target q, find the cheapest L and K.
2. Derive $C(q)$ and $MC(q)$.
3. Choose output, then calculate inputs and profit.

Useful starting points: $MP_L = q/(4L)$, $MP_K = q/(4K)$, and $LK = q^4$. First find the input ratio; then use the output target.

Solution: Inputs and Minimum Cost

1. Tangency:

$$\frac{MP_L}{MP_K} = \frac{K}{L} = \frac{w}{r} = 1 \quad \Longrightarrow \quad K = L.$$

2. Meet the target:

$$LK = q^4 \quad \Longrightarrow \quad L^2 = q^4 \quad \Longrightarrow \quad L^c(q) = K^c(q) = q^2.$$

3. Price the required inputs:

$$\boxed{C(q) = q^2 + q^2 = 2q^2}, \qquad \boxed{MC(q) = 4q}.$$

Doubling output requires four times as much of each input. The extra cost of another unit therefore rises with output.

Solution: Output, Inputs, and Profit

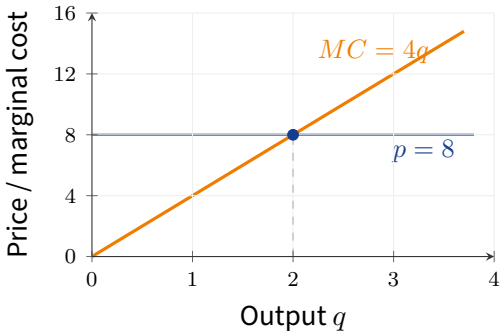
$$\pi(q) = 8q - 2q^2.$$

$$\pi'(q) = 8 - 4q = 0 \quad \Longrightarrow \quad q^* = 2.$$

Since $\pi''(q) = -4$, this is the maximum.

$$L^* = K^* = (q^*)^2 = 4.$$

$$\pi^* = 8(2) - (4 + 4) = 8 > \pi(0) = 0.$$



Use $p = MC$ to choose output; then return to conditional input demands to find the required input bundle.

The Same Problem in One Step

Instead of choosing output first, substitute the production function into revenue:

$$\max_{L,K \geq 0} \{pf(L,K) - wL - rK\}.$$

At a smooth interior optimum,

$$\boxed{pMP_L = w}, \qquad \boxed{pMP_K = r}.$$

The revenue from a little more of each input equals its cost.

Dividing the conditions gives $MP_L/MP_K = w/r$: the input mix is cost-minimizing.

Both methods solve the same problem. The two-step method separates **how to produce** from **how much to produce**, making costs and supply easier to understand.

Check: The One-Step Method

For $q = (LK)^{1/4}$, $p = 8$, and $w = r = 1$:

$$8\frac{q}{4L} = 1, \qquad 8\frac{q}{4K} = 1 \quad \Longrightarrow \quad L = K = 2q.$$

Use the production function to determine scale:

$$q^4 = LK = 4q^2 \quad \Longrightarrow \quad q = 2 \quad \text{at a positive optimum.}$$

Therefore $L = K = 4$ and profit is 8, exactly as before.

Dividing the input conditions gives only $K = L$. Keeping their **levels** brings the selling price into the output decision.

A Wage Change: Substitution and Scale

For a fixed output target: a higher wage changes the cheapest input mix. The firm substitutes away from labor along the same isoquant.

When output is also a choice: higher costs may make a smaller output worthwhile. The firm then moves to a different isoquant.

	What stays fixed?	What adjusts?
Conditional substitution	Output target	Input mix
Scale response	Output price, technology	Output and input needs

The consumer's income effect has no direct counterpart here: there is no fixed household budget. The firm's additional response comes through its **chosen scale of production**.

Now Restrict What the Firm Can Adjust

Until now, both labor and capital could adjust: the **long run**.

In the **short run**, capital is fixed at $\bar{K}$, while labor can adjust:

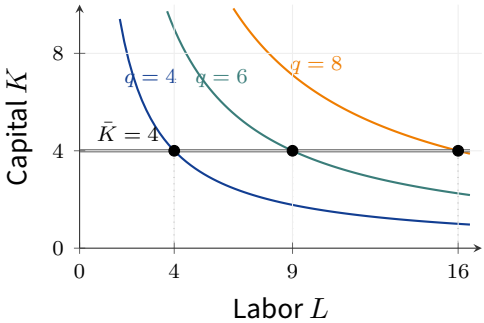
$$q = f(L, \bar{K}; A).$$

- The bakery can change staffing before it can change its ovens.
- It cannot choose a new labor–capital mix for each output target.
- "Long run" means enough time to adjust all modeled inputs, not a fixed number of months.

This changes the **constraints on the firm's choices**, not its objective or technology. More output now requires more labor on the same equipment.

For the next block, return to our benchmark $q = \sqrt{LK}$.

Fixed Capital Changes the Input Choice



The firm must stay on $K = \bar{K}$.

Each isoquant intersects that line at the labor needed for its output target:

$$q = \sqrt{L\bar{K}} \quad \Longrightarrow \quad L(q;\bar{K}) = \frac{q^2}{\bar{K}}.$$

At wage w, the labor bill is

$$VC(q) = w\frac{q^2}{\bar{K}}.$$

The firm can no longer move along an isoquant to find a tangency. With capital fixed, the output target determines required labor and its cost.

Total Product with Fixed Capital

With four units of capital,

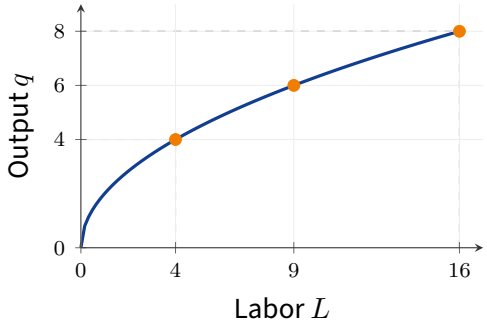
$$q = \sqrt{L\bar{K}} = 2\sqrt{L}.$$

More labor raises output, but each additional hour adds less than the previous hour.

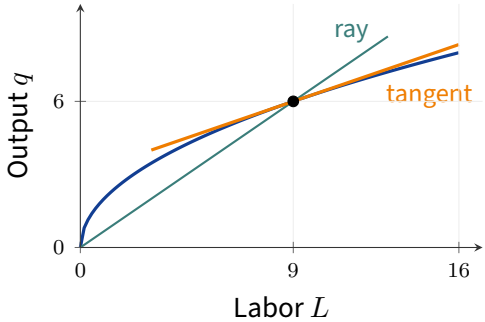
More workers share the same equipment: fixed capacity becomes a bottleneck.

Consequence: further output increases require progressively more labor.

Smaller additions to output do not mean that total output falls.



Two Slopes on the Production Curve



Average product is output per labor-hour: $\mathrm{AP}_L = q/L$.

At $L = 9$, output is $q = 6$.

Ray from the origin:

$$\mathrm{AP}_L = \frac{6}{9} = \frac{2}{3}.$$

Recall: marginal product is the tangent slope.

$$\mathrm{MP}_L = \frac{1}{\sqrt{9}} = \frac{1}{3}.$$

Here marginal product is below average product. Extra labor pulls the average down.

Exercise: Production with Fixed Capital

Keep $\bar{K} = 4$, wage $w = 1$, and technology $q = \sqrt{L\bar{K}} = 2\sqrt{L}$.

1. Compute output, average product, and marginal product at $L = 4$ and $L = 16$.
2. How much labor is required to produce $q = 6$? What is its cost?
3. Derive required labor $L(q; \bar{K})$ and variable cost $VC(q)$. Why does doubling output more than double the labor bill?

$$AP_L = \frac{q}{L}, \qquad MP_L = \frac{dq}{dL}.$$

For the output target, solve the production constraint for L. Capital is already fixed, so there is no input-mix tangency to solve.

Solution: Productivity and Labor

$$q = 2\sqrt{L}, \qquad AP_L = \frac{2}{\sqrt{L}}, \qquad MP_L = \frac{1}{\sqrt{L}}.$$

Labor L	Output q	AP_L	MP_L
4	4	1	$1/2$
16	8	$1/2$	$1/4$

To produce 6 units: $6 = 2\sqrt{L}$, hence $L = 9$ and $wL = 9$. More generally,

$$\boxed{L^c(q;\bar{K}) = q^2/\bar{K}}, \qquad \boxed{VC(q) = wq^2/\bar{K}}.$$

Doubling output from 4 to 8 needs four times the labor because capital cannot expand with it. This comes from the **production constraint**.

From Inputs to Short-Run Costs

We know the least labor needed for each target under $q = \sqrt{L\bar{K}}$:

$$L^c(q;\bar{K}) = \frac{q^2}{\bar{K}}.$$

Now attach prices. Assume the capital payment $r\bar{K}$ is committed for the period:

$$C_{SR}(q;\bar{K}) = \underbrace{r\bar{K}}_{F} + \underbrace{w\frac{q^2}{\bar{K}}}_{VC(q)}.$$

- More output requires more paid labor: this cost varies with q.
- The existing capital payment is due even at $q = 0$.

A fixed input constrains production. An unavoidable payment affects profit but does not change the cost of producing **one more unit**.

Average Costs and Marginal Cost

Measure	Definition	Economic question
Average fixed cost	$\mathrm{AFC} = F/q$	Fixed cost per unit?
Average variable cost	$\mathrm{AVC} = VC(q)/q$	Variable cost per unit?
Average total cost	$\mathrm{AC} = C(q)/q$	Total cost per unit?
Marginal cost	$\mathrm{MC} = C'(q)$	Cost of a little more output?

$$\mathrm{AC} = \mathrm{AFC} + \mathrm{AVC}, \qquad \mathrm{MC} = VC'(q).$$

Average cost spreads total cost over output. Marginal cost measures the cost of expanding production.

Calculate Every Cost Measure

Continue with $q = \sqrt{LK}$, $\bar{K} = 4$, and $w = r = 1$:

$$F = 4, \quad VC(q) = q^2/4, \quad C_{SR}(q) = 4 + q^2/4.$$

Divide by output to obtain averages; differentiate to obtain marginal cost:

$$AFC = \frac{4}{q}, \quad AVC = \frac{q}{4}, \quad AC = \frac{4}{q} + \frac{q}{4}, \quad MC = \frac{q}{2}.$$

q	Required L	Total cost	AFC	AVC	AC	MC
4	4	8	1	1	2	2
8	16	20	0.5	2	2.5	4

At $q = 8$, $AP_L = 8/16 = 1/2$, so $w/AP_L = 2 = AVC$. Also $MP_L = 1/4$, so $w/MP_L = 4 = MC$.

Productivity and Marginal Cost

Suppose labor is the only variable input and its wage is constant.

$$VC(q) = wL(q; \bar{K}).$$

An extra unit of output requires $1/\mathrm{MP}_L$ additional labor units:

$$\boxed{\mathrm{MC} = \frac{w}{\mathrm{MP}_L}}, \qquad \boxed{\mathrm{AVC} = \frac{w}{\mathrm{AP}_L}}.$$

- When an extra labor-hour produces less, each extra unit costs more.
- When output per labor-hour is high, variable cost per unit is low.

With $w = 20$, a marginal product of 10 gives $\mathrm{MC} = 2$. If marginal product falls to 5, MC rises to 4.

Capital and Labor Productivity

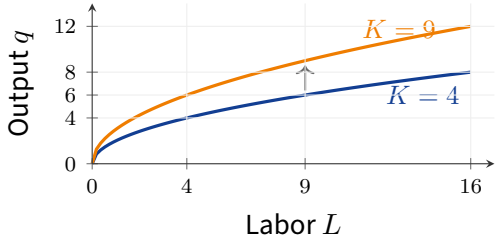
For $q = \sqrt{LK}$,

$$AP_L = \sqrt{K/L}, \qquad AP_K = \sqrt{L/K}.$$

With the same technology and labor:

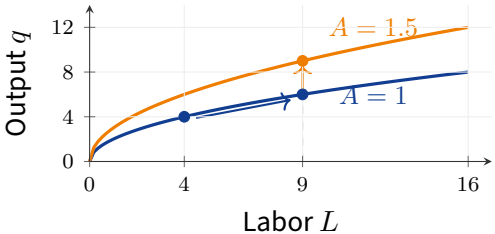
L	K	q	AP_L	AP_K
4	4	4	1	1
4	9	6	1.5	2/3

More capital raises AP_L and $MP_L = AP_L/2$, but lowers AP_K.



Higher output per worker can reflect more capital per worker. Neither input ratio alone identifies better technology.

Technological Progress



$$q = A\sqrt{LK}, \qquad K = 4.$$

Movement along the curve: more labor, same capital and technology.

Shift of the curve: more output from the same inputs.

Better methods, software, or organization can change productive possibilities.

Compare two bakeries: A produces 100 with $L = 10$, $K = 4$; B produces 160 with $L = 10$, $K = 16$. Does B's higher output per worker identify better technology? Do these observations identify MP_L?

Why Cost Curves Can Look Different

The square-root example has diminishing marginal product from the start:

$$AVC = q/4 \quad \text{and} \quad MC = q/2$$

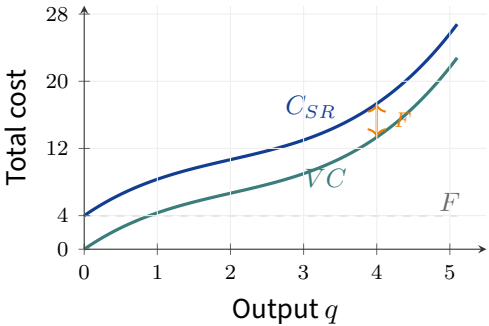
are both increasing. Its $AC = 4/q + q/4$ is U-shaped.

Some technologies allow initial productivity gains, followed by diminishing productivity as fixed inputs become bottlenecks.

The next diagrams illustrate this possibility: AVC and MC can initially fall and then rise. The illustrative formula is in the appendix.

The definitions and decision rules are unchanged. The precise curve shapes depend on technology; U-shaped AVC is not a universal law.

Fixed and Variable Costs



With capacity fixed,

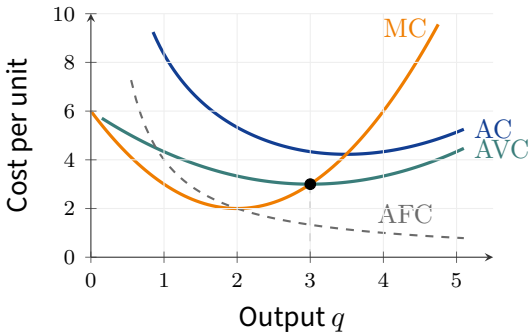
$$C_{SR}(q) = F + VC(q).$$

- F does not vary with current output.
- $VC(q)$ increases as the firm uses more variable inputs.

For the shutdown analysis, assume F is unavoidable over the current horizon and $VC(0) = 0$.

Total and variable cost differ by the same amount F at every output.

The Usual Short-Run Cost Curves



- AFC falls as fixed cost spreads across more units.
- AVC can fall initially, then rise as variable inputs become less productive.
- AC combines these forces.
- Rising MC crosses AVC and AC at their respective minima.

The gap $\mathrm{AC} - \mathrm{AVC} = F/q$ narrows as output rises.

A common configuration, not a universal shape. Technology and the adjustment horizon determine the curves.

Marginal Cost Moves Average Cost

Suppose average cost is currently 10 per unit.

- An extra unit costing 6 pulls the average down.
- An extra unit costing 14 pushes the average up.

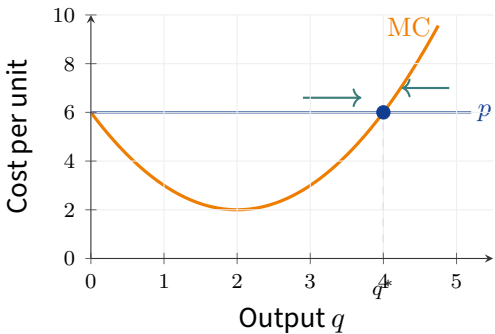
The equation captures exactly this reasoning:

$$\frac{d\mathrm{AC}}{dq} = \frac{qC'(q) - C(q)}{q^2} = \frac{\mathrm{MC} - \mathrm{AC}}{q}.$$

At a smooth interior minimum of average cost, $\mathrm{MC} = \mathrm{AC}$. The same argument gives $\mathrm{MC} = \mathrm{AVC}$ at a smooth interior minimum of AVC.

This is the same marginal-versus-average logic we used for labor productivity.

Choosing Output



At the chosen output,

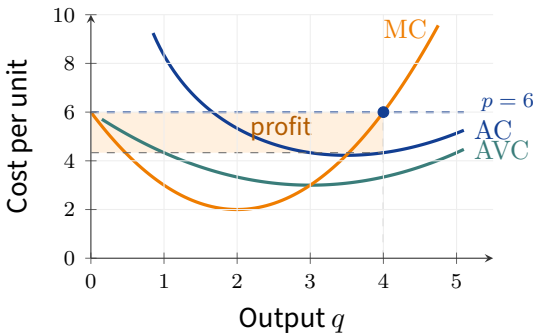
$$p = \mathrm{MC}(q^*).$$

On the rising part of MC:

- just below q^*, an extra unit is profitable;
- just above q^*, it costs more than it earns.

The relevant candidate is a **profit maximum**, not every intersection of price and marginal cost.

Average Cost and Profit



Profit equals output times the profit margin:

$$\pi(q^*) = [p - \mathrm{AC}(q^*)]q^*.$$

- MC helps choose output.
- AC tells us whether revenue covers total cost at that output.

The shaded rectangle has width q^* and height $p - \mathrm{AC}(q^*)$.

The firm need not produce at minimum AC.

Fixed Costs and Sunk Costs

Fixed cost	Does not vary with the current quantity produced.
Sunk cost	Cannot be recovered or avoided by changing the decision now.

- A nonrefundable past setup payment is sunk.
- A license fee paid only if the firm operates is fixed but avoidable.
- A machine the firm can sell has an opportunity cost even if it already owns it.

For shutdown or exit, compare revenue with **all costs that the decision can avoid**. The AVC rule assumes that all fixed costs are unavoidable in the short run.

The Shutdown Decision

Suppose fixed cost F must be paid even if current production stops.

$$\pi(q) = pq - VC(q) - F, \qquad \pi(0) = -F.$$

Producing a positive quantity beats shutting down if

$$pq - VC(q) \geq 0 \quad \Longleftrightarrow \quad p \geq \mathrm{AVC}(q).$$

The firm produces when some output covers its **avoidable cost**. Under these assumptions, the shutdown price is $\min_{q>0} \mathrm{AVC}(q)$.

A firm can rationally keep producing even while making an economic loss.

Producing at a Loss

Suppose the best positive-output choice is $q = 10$ at price $p = 10$.

	Produce	Shut down
Revenue	100	0
Variable cost	80	0
Unavoidable fixed cost	50	50
Profit	-30	-50

Here $\text{AVC} = 8 < p = 10 < \text{AC} = 13$.

Production contributes 20 toward a cost the firm must pay anyway. Shutting down would increase the loss.

Produce Now, but Do Not Renew

Continue with the same firm. Next period, its best production plan would again generate revenue of 100 and variable cost of 80.

The next fixed payment of 50 is **not yet committed**: the firm can avoid it by closing before renewal.

Next period	Renew and produce	Close before renewal
Revenue	100	0
Variable cost	80	0
Avoidable fixed payment	50	0
Profit	-30	0

- **Current period:** produce, because $-30 > -50$. The committed payment is unavoidable.
- **At renewal:** close, because $0 > -30$. The next payment is avoidable.

Three Production Decisions

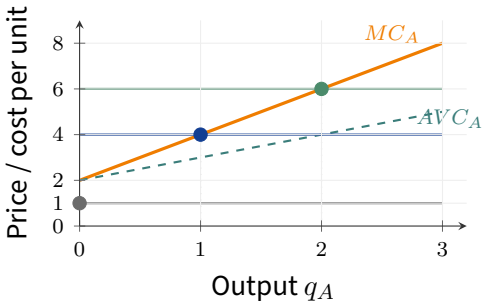
First find the best positive-output candidate on rising MC, where $p = MC$. Then compare the alternatives:

At the candidate q^*	Current decision	Why?
$p > AC(q^*)$	Produce at a profit	Revenue covers total cost.
$AVC(q^*) < p < AC(q^*)$	Produce at a loss	Production covers variable cost and part of the unavoidable bill.
$p < \min AVC$	Shut down	Every positive quantity adds to the unavoidable loss.

Use MC to choose **how much**, AVC to decide **whether to produce now**, and AC to measure **profit or loss**.

At $p = AC(q^*)$, economic profit is zero. At $p = \min AVC$, shutdown and a positive quantity tie.

Prices and Production Choices



New example:

$$C_A(q) = F_A + 2q + q^2.$$

F_A is unavoidable. Here $MC_A = 2 + 2q$ and $AVC_A = 2 + q$.

Price	Best output
$p = 1$	$q_A = 0$: shut down
$p = 4$	$q_A = 1$: $p = MC_A$
$p = 6$	$q_A = 2$: $p = MC_A$

At $p \leq 2$, the firm shuts down.

As in Unit 1, change the **price**, hold the other conditions fixed, and record the agent's **optimal quantity**.

From Choices to Individual Supply

At each possible price, the firm solves

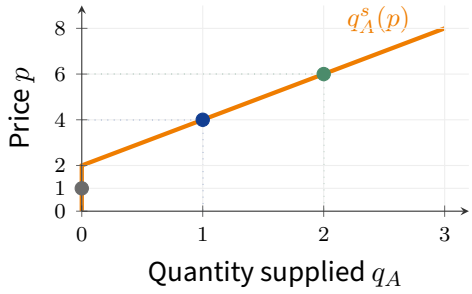
$$\max_{q \geq 0} \{pq - C_A(q)\}.$$

For our example, its choices are

$$q_A^s(p) = \begin{cases} 0, & p \leq 2, \\ (p-2)/2, & p > 2. \end{cases}$$

Supply is the whole schedule, not one observed quantity.

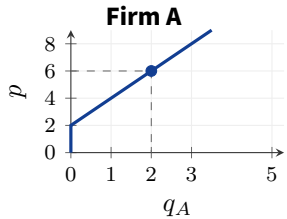
A price change gives a **movement along** the curve.

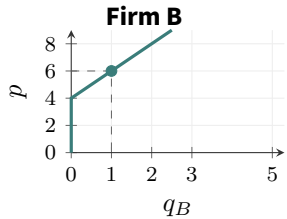


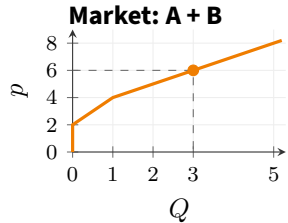
Unit 1: optimal consumption at each price gives individual demand.
Unit 3: optimal production at each price gives individual supply.

From Individual to Market Supply

Firm A has $MC_A = 2 + 2q_A$; Firm B has $MC_B = 4 + 2q_B$. Both have unavoidable fixed costs and $VC_i(0) = 0$.







$$Q^s(p) = q_A^s(p) + q_B^s(p) \quad \Longrightarrow \quad Q^s(6) = 2 + 1 = 3.$$

Horizontal summation, just as with demand: fix one common price and add quantities. Between $p = 2$ and $p = 4$, only A produces; above $p = 4$, both do.

Short run: the set of firms and their installed capital are fixed; input prices are held constant.

What Shifts the Firm's Supply Curve?

Supply depends on costs, not only on the selling price.

Change	Implication, other conditions fixed
Higher output price	Movement along the existing supply curve.
Higher variable-input price	Can raise MC and reduce supply at a given price.
Higher unavoidable fixed cost	Raises AC, with no change in MC, AVC, or short-run supply.
Improved technology	Weakly lowers total cost at each output. The effect on MC depends on the improvement.

To predict supply, identify **which costs change** and **which inputs can adjust**.

Long-Run Cost: The Value of Flexibility

With a fixed plant $\bar{K}$, the firm chooses only variable inputs:

$$C_{SR}(q;\bar{K}) = \min_{L\geq 0}\{wL + r\bar{K} :\ f(L,\bar{K}) \geq q\}.$$

In the long run, it also chooses the plant:

$$C_{LR}(q) = \min_{K\geq 0} C_{SR}(q;K).$$

Therefore, for every feasible output and any given plant,

$$C_{LR}(q) \leq C_{SR}(q;\bar{K}).$$

More flexibility cannot raise the minimum cost of a **given output**, at the same input prices and with the same technology.

Same Technology, Different Horizons

Let $q = \sqrt{LK}, w = r = 1$.

Short run: $\bar{K} = 4$**.**

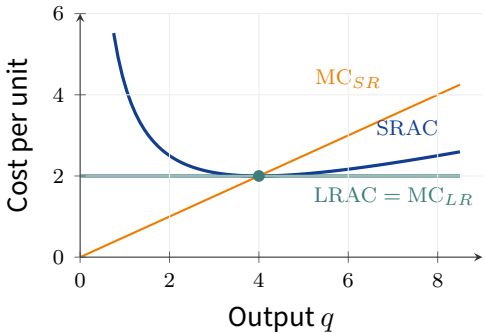
$$L = q^2/4, \quad C_{SR} = 4 + q^2/4.$$

$$\text{SRAC} = 4/q + q/4, \quad \text{MC}_{SR} = q/2.$$

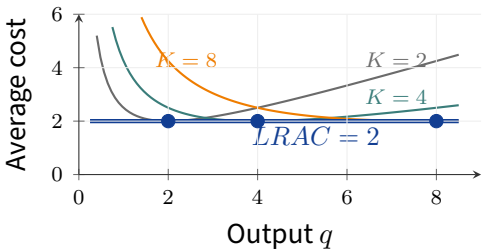
Long run: both inputs adjust.

$$L^c = K^c = q, \quad C_{LR} = 2q.$$

$$\text{LRAC} = \text{MC}_{LR} = 2.$$



Choosing the Cheapest Plant at Each Output



Keep $q = \sqrt{LK}$ and $w = r = 1$.

$$SRAC(q; K) = \frac{K}{q} + \frac{q}{K}.$$

Choose the cheapest plant at each output:

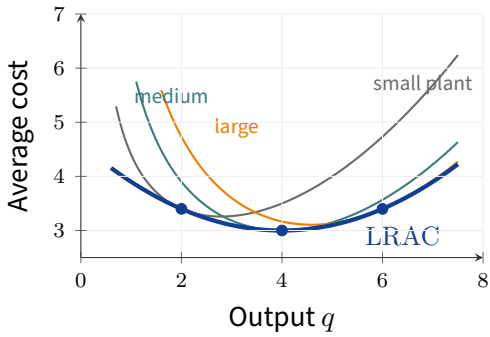
$$LRAC(q) = \min_{K>0} SRAC(q; K) = 2.$$

The best choice is $K = q$.

The **lower envelope** selects a different plant for each output. It is flat here because technology has **constant returns to scale**.

Three plants are shown; the envelope allows every positive capital choice.

Long-Run Average Cost



A different illustrative technology: here the envelope is U-shaped. Each short-run curve represents a plant size.

The same envelope principle selects the cheapest plant at each output.

$$\mathrm{LRAC}(q) = \min_{K} \mathrm{SRAC}(q; K).$$

The curves touch where a plant is the best choice for that output.

The contact need not occur at that plant's minimum average cost.

Three selected plant sizes are shown. The smooth envelope allows a continuous choice of plant size.

Economies of Scale and Long-Run Cost

Economies of scale: long-run average cost falls as output expands.

- Specialization or indivisible equipment can make larger scale cheaper.
- Coordination difficulties or scarce managerial capacity can work in the opposite direction.

With constant input prices, suppose $f(tL, tK) = t^s f(L, K)$. Then

$$C_{LR}(q) \propto q^{1/s}, \qquad \mathrm{LRAC}(q) \propto q^{1/s-1}.$$

Increasing returns, $s > 1$	Constant returns, $s = 1$	Decreasing returns, $s < 1$
Falling LRAC	Flat LRAC	Rising LRAC

A U shape is possible, but is not implied by the definition of the long run.

Shutdown versus Exit

In the long run, the firm can release its inputs and avoid future operating costs.

$$\max_{q \geq 0} \{pq - C_{LR}(q)\}, \qquad C_{LR}(0) = 0.$$

A positive output is worthwhile if some quantity covers its full economic cost:

$$p \geq \min_{q>0} \mathrm{LRAC}(q).$$

- Short-run shutdown asks whether current production covers avoidable costs.
- Long-run exit asks whether continued operation covers all future economic costs.

Zero economic profit already compensates the resources the owner supplies at their opportunity costs.

Checkpoint: Changes in Costs

For each scenario, explain what changes and why.

1. The firm must pay a higher rent on its existing factory this month, whether it produces or not. Does MC change? Does AC change?
2. A more efficient method raises output from every input bundle. What happens to minimum cost at a fixed output?
3. The firm can now choose its plant size. Can its minimum cost at a given output be higher than with its old plant?

Separate a change in fixed expenditure, a change in productive possibilities, and a change in the choices available.

Reading Production Data

A data set records the firm's chosen inputs, output, and spending.

- **Output per worker is an average.** It depends on capital, technology, and product quality.
- **Input choices respond to productivity.** More productive firms may hire more workers, so a cross-firm correlation need not reveal a marginal product.
- **Revenue combines quantity and price.** Higher sales per worker can reflect higher prices rather than more physical output.

Data-science lesson

Before estimating a production relationship, define the output and its quality, identify the inputs that can adjust, and state the counterfactual you want to learn.

Notebook 2: Return to the Same Plants

Sections 4–11 | Estimate, check, explain

Resume at RETURN AFTER THEORY, Section 4.

1. Estimate fuel requirements from **2021 plant histories**.
2. Turn technology and input prices into costs.
3. Predict daily output in **2022** and renewal for **2023**.
4. Change selling prices and the emissions charge separately.

The theory supplies the decision rules. The data supply the technology estimates, prices, capacity and commitments.

A Production Model for These Plants

Equipment is fixed. Fuel is adjustable.

$$B_i(q) = a_i q + b_i q^2, \quad a_i, b_i > 0.$$

$$0 \le q \le K_i.$$

$B_i(q)$ is the least fuel needed for q MWh; K_i is the maximum MWh per day.

Cost minimization means using exactly this much fuel.

From the data

Output q: `output_mwh`
Fuel B: `fuel_gj`
Capacity K: `capacity_mwh`

Estimate a_i, b_i from each plant's output and fuel history.

This is the fixed-equipment production relationship written as **input required for output**. There is only one adjustable input here.

From Fuel to Variable Cost

Fuel costs f_t EUR/GJ. Each GJ emits z_i kg, charged at τ_t EUR/kg.

$$VC_{it}(q) = \underbrace{f_t B_i(q)}_{\text{fuel bill}} + \underbrace{\tau_t z_i B_i(q)}_{\text{emissions bill}} + \underbrace{v_i q}_{\text{other expenses}}.$$

Write the effective fuel price as $s_{it} = f_t + \tau_t z_i$. Then

$$VC_{it}(q) = s_{it}(a_i q + b_i q^2) + v_i q.$$

100 GJ $\times$ 0.1 kg/GJ $=$ 10 kg. At EUR 2.50/kg, the emissions bill is EUR 25. The charge therefore multiplies **fuel use**.

Which Costs Can the Plant Avoid?

Cost or constraint	Today	At renewal
Fuel, emissions, other variable bills	Choose through output	Choose through output
Annual commitment F_i	Already unavoidable	Avoid by closing
Past construction H_i	Unrecoverable	Still unrecoverable
Existing capacity K_i	Fixed output limit	Same plant size

F_i is committed for this year, but the **next** annual bill is avoidable. Construction spending H_i is sunk at both decision dates.

Renewal chooses whether to keep this plant; it does not choose a new capacity.

Daily Choice: How Much to Produce?

With the annual bill already committed, choose

$$\max_{0 \le q \le K_i} \{p_t q - VC_{it}(q)\}, \qquad MC_{it}(q) = s_{it}(a_i + 2b_i q) + v_i.$$

Price relative to marginal cost	Best daily choice
$p_t \le MC_{it}(0)$	Idle: $q^* = 0$
$MC_{it}(0) < p_t < MC_{it}(K_i)$	Choose $p_t = MC_{it}(q^*)$
$p_t \ge MC_{it}(K_i)$	Produce at capacity: $q^* = K_i$

Neither the committed annual bill nor past construction spending changes the extra earnings from another MWh.

Annual Choice: Renew or Close?

At renewal, choose whether to open ($r_i = 1$) and daily quantities:

$$\max_{\substack{r_i \in \{0,1\} \\ 0 \le q_{it} \le r_i K_i}} \left\{ \sum_{t \text{ in next year}} [p_t q_{it} - VC_{it}(q_{it})] - r_i F_i \right\}.$$

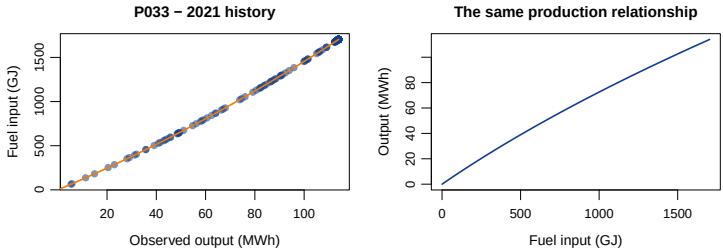
If it opens, let q_{it}^* be its best daily output and calculate

$$S_i = \sum_t [p_t q_{it}^* - VC_{it}(q_{it}^*)].$$

Renew if $S_i \ge F_i$. Closing gives zero future profit; renewal gives $S_i - F_i$. Past construction expenditure is absent from both alternatives.

For January 2023, the known daily price schedule repeats 2022; costs and policy persist.

Estimate Each Plant's Technology



$$\widehat{B}_i(q) = \widehat{a}_i q + \widehat{b}_i q^2.$$

Fit each plant's 2021 fuel–output curve. Hold back 2022 to check whether the decision model travels to a new price and policy environment.

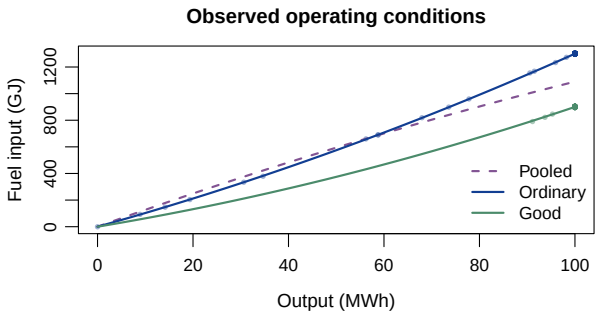
Fuel Productivity Without Innovation

Estimated technology	P007	P030	P033
$\widehat{a}_i$	10.14	12.75	11.75
$\widehat{b}_i$	0.0257	0.0450	0.0280

$$\frac{B_i(q)}{q} = a_i + b_i q, \qquad AP_B = \frac{1}{a_i + b_i q}, \qquad MP_B = \frac{1}{a_i + 2b_i q}.$$

At fixed equipment, reducing output lowers fuel per MWh and raises average fuel productivity. The plant moves along the **same** production relationship.

When Efficiency Changes with Output



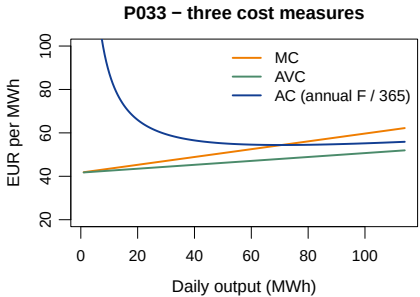
On good-condition days a plant uses less fuel *and* chooses more output.

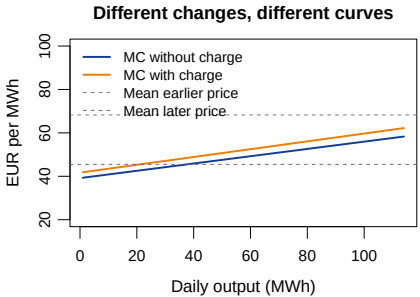
Pooling conditions estimates $\widehat{b} = -0.0196$.

Allowing for observed condition gives $\widehat{b} = 0.0302$; the true value is 0.03.

The simple fix here is to fit curves by **observed condition**. If efficiency is hidden, this control is unavailable: more identification is needed.

Separate Price and Cost Changes





Lower selling prices move the revenue line down. The emissions charge raises MC and AVC. A higher committed F_i raises AC, but leaves MC unchanged.

Apply the Daily Decision Rule

Solve $p_t = MC_{it}(q)$, then respect the output limits:

$$\widehat{q}_{it} = \min\left\{K_i,\ \max\left\{0, \frac{p_t - v_i - s_{it}\widehat{a}_i}{2s_{it}\widehat{b}_i}\right\}\right\}.$$

- Negative interior solution? Choose zero.
- Interior solution above capacity? Choose K_i.
- Otherwise, produce the interior quantity.

The theory turns an estimated fuel curve into a prediction for **every plant-day**. Annual fixed costs do not enter this daily rule.

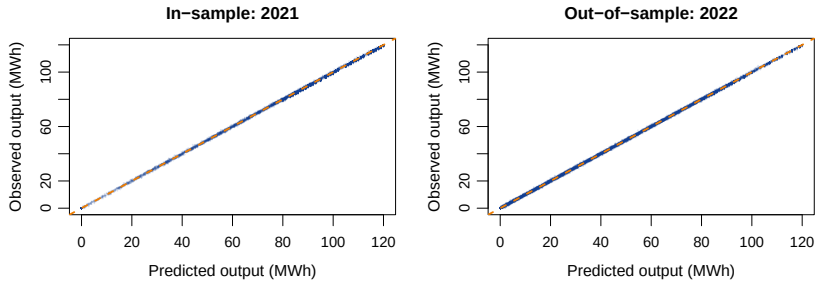
P033: Producing Loses Less Than Idling

One day (EUR unless stated)	Produce	Idle
Output (MWh)	22.9	0
Revenue	1,048.6	0
Variable cost	1,001.5	0
Allocated commitment $F_i/365$	449.6	449.6
Profit after allocation	-402.4	-449.6

Producing earns EUR 47.1 towards an unavoidable bill. A negative accounting profit is not enough to justify shutting down today.

Check the Daily Output Predictions

Section 7 | Validate before counterfactuals



Prediction check	2021: estimation year	2022: held back
Mean absolute error (MWh)	0.287	0.286
Operating-status accuracy	99.98%	99.83%

Check Shutdowns as Well as Quantities

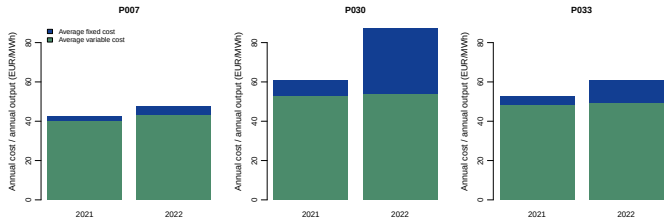
All 43,800 plant-days	Predicted idle	Predicted producing
Observed idle	7,524	41
Observed producing	0	36,235

- A high operating rate can make status accuracy look good.
- Inspect both prediction errors and the idle/producing comparison.
- The held-back year tests the model under different economic conditions.

These checks support using the model in this synthetic exercise. They do not establish that every real plant faces the same constraints.

Return to the Average-Cost Puzzle

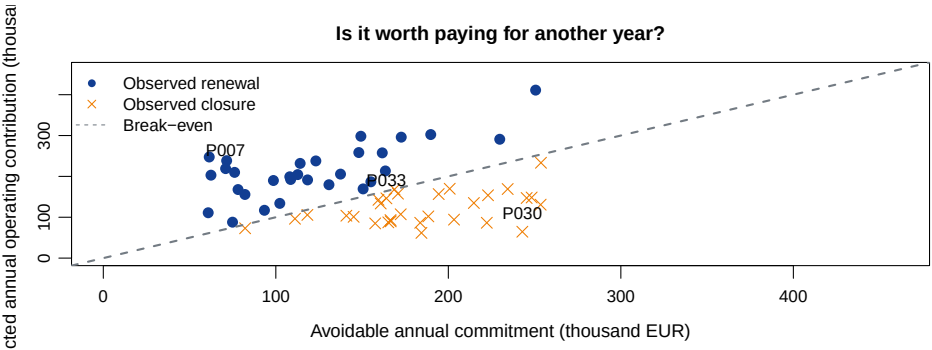
Section 8 | Explain rising average cost



$$\text{Annual } AC_i = \frac{\sum_t VC_{it}}{Q_i} + \frac{F_i}{Q_i}.$$

Less output spreads the same annual bill over fewer MWh. The charge raises the cost of each GJ. Fuel per MWh can fall while total cost per MWh rises.

Check the Annual Renewal Decision



Above the diagonal, the next year covers its commitment. Below it, the plant closes. The model predicts all 30 observed nonrenewals.

Reconcile Daily Output and Renewal

Predicted next year (EUR thousand)	P007	P030	P033
Best operating contribution S_i	219.0	64.6	146.3
Avoidable commitment F_i	70.9	242.9	164.1
Profit from renewal $S_i - F_i$	148.1	-178.3	-17.8
Renew?	Yes	No	No

P033 can cover variable costs on many days, yet cannot cover the next annual commitment. Operating during 2022 and closing for 2023 are consistent.

Near the boundary, be cautious: P040's fitted annual profit is only −EUR 503.

Separate Prices and Policy

Counterfactual	Electricity prices	Emissions charge
Earlier conditions	2021 daily schedule	Zero
Lower prices only	2022 daily schedule	Zero
Policy only	2021 daily schedule	EUR 2.50/kg
Both changes	2022 daily schedule	EUR 2.50/kg

In every scenario, start with the same 60 plants and re-optimize daily output. Then compare annual contribution with the next commitment.

Keep technologies, capacity, fuel prices and annual commitments fixed. Allow **quantities and renewal decisions** to respond.

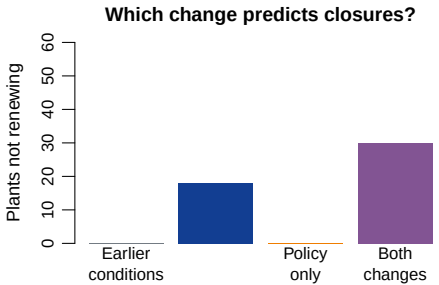
Compare Profits Across Scenarios

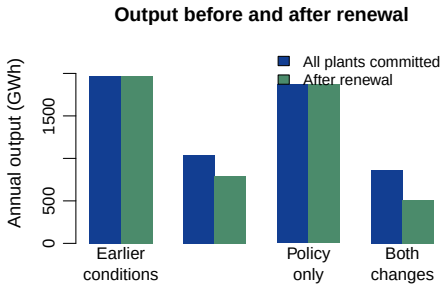
Scenario	Profit if all commit	Plants not renewing
Earlier conditions	40.11 million	0
Lower prices only	4.03 million	18
Policy only	33.28 million	0
Both changes	0.84 million	30

Profits in EUR; each scenario lets daily production adjust.

At later prices, adding the charge lowers total profit from EUR 4.03m to EUR 0.84m. Removing it raises profits, but does not make every plant viable.

Output Before and After Renewal





With both changes, output is 861 GWh while commitments remain in place, and 513 GWh after nonrenewing plants leave.

Revisit the Same Three Plants

Annual profit (EUR thousand)	Earlier	Prices only	Policy only	Both
P007	866.3	226.4	734.7	148.1
P030	362.5	-125.6	188.8	-178.3
P033	683.5	30.8	569.4	-17.8

- **P007** remains viable in all four environments.
- **P030** closes even without the charge at later prices.
- **P033** crosses the renewal threshold only with both changes.

A policy can be decisive for some firms even when lower prices account for most of the aggregate profit decline.

Answer the Original Claim

Sections 10–11 | Return to the initial memo

At the later selling prices:

Model prediction	Plants
Close with the charge	30
Still close without the charge	18
Avoid closure when the charge is removed	12

Removing the charge prevents **12 of 30** predicted closures in this synthetic fleet. The other **18** remain unprofitable at the later selling prices.

Prices and policy interact with each plant's renewal threshold. This is a conditional comparison, not a unique division of blame.

How This Connects to the Paper

Linn and McCormack (2019) study how energy-market changes and environmental regulation affect coal-plant profits and retirements.

Shared empirical logic

- Model plant operation and continuation.
- Check predictions against observed choices.
- Compare separate market and policy counterfactuals.

Our classroom simplification

- Invented electricity-price decline and emissions charge.
- Fixed equipment and known future prices.
- No electricity-market equilibrium or investment.

The paper attributes nearly all profitability declines and retirements to electricity consumption and natural-gas prices. Our numbers are **not** its estimates.

What Did Production Theory Add?

In the raw data	What the theory lets us do
Fuel and output move together	Estimate technology under explicit assumptions
Fuel per MWh falls	Separate a movement along a curve from innovation
Average cost rises	Distinguish fixed-cost dilution from marginal cost
Loss-making plants produce	Compare avoidable costs for the relevant decision
Two shocks coincide	Predict choices when each shock changes separately

We used a model of choices to build and check counterfactual predictions. The answer depends on the model as well as the observations.

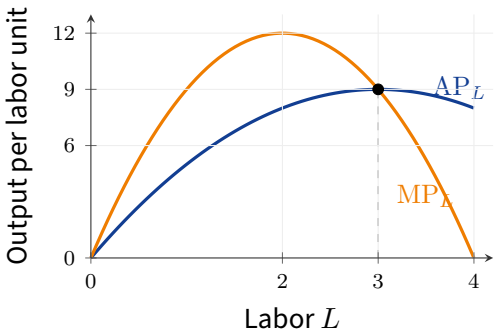
Demand Meets Supply in Unit 4

Technology	Which input bundles can produce each output?
Cost minimization	Which of those bundles is cheapest at current input prices?
Profit maximization	Which output gives the highest revenue net of cost?
Firm supply	How does that choice change with the output price?

Unit 1 gave us household demand. Unit 3 gives us firm supply.

In Unit 4, the competitive price makes the quantity households want to buy equal the quantity firms want to sell.

Average and Marginal Product Curves



If the next unit of labor adds:

- more than the average, AP_L rises;
- less than the average, AP_L falls.

At a smooth interior peak of AP_L,

$$\mathrm{MP}_L = \mathrm{AP}_L.$$

Marginal product can already be falling while average product is still rising.

Illustrative technology: $q = 6L^2 - L^3$ for $0 \le L \le 4$, with capital fixed.

Appendix: Average and Marginal Product

With capital fixed, write $q = f(L)$ and $L > 0$.

$$\mathrm{AP}_L = \frac{f(L)}{L}.$$

Differentiating,

$$\frac{d\mathrm{AP}_L}{dL} = \frac{Lf'(L) - f(L)}{L^2} = \frac{\mathrm{MP}_L - \mathrm{AP}_L}{L}.$$

- If $\mathrm{MP}_L > \mathrm{AP}_L$, average product rises.
- If $\mathrm{MP}_L < \mathrm{AP}_L$, average product falls.
- At a smooth interior maximum of average product, they are equal.

Appendix: Cobb–Douglas Production

$$q = AL^{\alpha}K^{\beta}, \qquad A, \alpha, \beta > 0.$$

$$\mathrm{MP}_L = \alpha\frac{q}{L}, \qquad \mathrm{MP}_K = \beta\frac{q}{K}, \qquad \mathrm{MRTS}_{LK} = \frac{\alpha}{\beta}\frac{K}{L}.$$

- α is the labor elasticity: $\partial \log q / \partial \log L = \alpha$.
- $0 < \alpha < 1$ gives diminishing marginal product of labor.
- Returns to scale depend on $\alpha + \beta$.

$$\log q = \log A + \alpha \log L + \beta \log K.$$

The log equation suggests a regression, but interpreting its coefficients as technology requires a strategy for endogenous input choices and unobserved productivity.

Appendix: Corners and the Lagrangian

For differentiable technology, include nonnegative inputs explicitly:

$$\mathcal{L} = wL + rK + \mu[q - f(L, K)] - \eta_L L - \eta_K K.$$

At a regular optimum,

$$w - \mu \mathrm{MP}_L = \eta_L \geq 0, \qquad r - \mu \mathrm{MP}_K = \eta_K \geq 0,$$

$$\eta_L L = \eta_K K = 0, \qquad \mu \geq 0, \qquad \mu[q - f(L, K)] = 0.$$

- An input used in positive quantity satisfies the corresponding equality $w = \mu \mathrm{MP}_L$ or $r = \mu \mathrm{MP}_K$.
- An unused input may cost more than the output it would add is worth at the shadow value μ.

At kinks, use the geometry or appropriate nonsmooth conditions.

Appendix: Cost and Profit

The production and cost descriptions encode the same feasible technology under suitable regularity conditions.

$$C(q; w, r) = \min_{f(L,K) \geq q} \{wL + rK\}.$$

Profit maximization can then be written as

$$\pi(p, w, r) = \max_{q \geq 0} \{pq - C(q; w, r)\}.$$

At a smooth interior optimum, $\mu = C_q = p$, so

$$p\mathrm{MP}_L = w, \qquad p\mathrm{MP}_K = r.$$

An input's extra revenue equals its extra cost. The cost-minimization conditions and the output-choice condition together recover the direct profit-maximization conditions.

Appendix: The Illustrative Cost Function

The short-run diagrams use one illustrative cost function:

$$C(q) = 4 + 6q - 2q^2 + \frac{q^3}{3}, \qquad q \geq 0.$$

$$F = 4, \qquad \mathrm{AVC} = 6 - 2q + \frac{q^2}{3}, \qquad \mathrm{MC} = 6 - 4q + q^2.$$

- $\mathrm{MC} = (q-2)^2 + 2 > 0$: total cost always rises.
- $\min \mathrm{AVC} = 3$ at $q = 3$: this is the shutdown price.
- At $p = 6$, the firm chooses $q = 4$ and earns $20/3$.

$$q^s(p) = \begin{cases} 0, & p < 3, \\ \{0, 3\}, & p = 3, \\ 2 + \sqrt{p-2}, & p > 3. \end{cases}$$

Appendix: Checkpoint Answers

Higher labor productivity

- Bakery B has $\mathrm{AP}_L = 160/10 = 16$, compared with 10 for A.
- B also uses more capital. These data cannot establish better technology.
- A marginal product requires a change in labor holding other inputs and technology fixed. The two observations do not provide that comparison.

Changes in cost

- A higher unavoidable fixed payment raises AC by $\Delta F/q$, leaving MC and AVC unchanged.
- An expanded production set weakly reduces minimum cost at each output.
- Allowing the firm to change plant size weakly reduces minimum cost because it can still choose the old plant.

References and Further Reading

Base theory

Espinola-Arredondo, Ana, and Felix Munoz-Garcia (2020).
Intermediate Microeconomic Theory: Tools and Step-by-Step Examples. MIT Press.

- Chapter 7: production functions.
- Chapter 8: cost minimization.
- Chapter 9: competitive firms and equilibrium.

Optional: Fixed Costs or Cheaper Fuel?

Optional extension A | Notebook question 14

Support	Current output	Closures	Output after renewal
No support	861	30	513
20% lower annual fixed cost	861	19	658
10% lower private fuel price	1,078	16	855

Output in GWh; later selling prices and emissions charge in all scenarios.

- A smaller future annual commitment encourages renewal.
- Cheaper fuel lowers MC, changing current output and renewal.

The two interventions have different budgets. This compares mechanisms, not welfare or cost-effectiveness.

Optional: How Robust Is the Answer?

Optional extension B | Notebook question 15

Fuel requirement	Annual bill -10%	Annual bill unchanged	Annual bill $+10\%$
-5%	12	8	10
Unchanged	12	12	7
$+5\%$	9	9	9

Entries: closures prevented by removing the charge at later selling prices.

Across these nine assumptions, predicted closures with both changes range from **18 to 35**; removing the policy prevents **7 to 12**.

These are transparent sensitivity checks, not confidence intervals. Near the renewal threshold, small cost differences can change a plant's decision.

Application Reference and Reproducibility

Joshua Linn and Kristen McCormack (2019). “The Roles of Energy Markets and Environmental Regulation in Reducing Coal-Fired Plant Profits and Electricity Sector Emissions.” *The RAND Journal of Economics*, 50(4), 733–767.

doi:10.1111/1756-2171.12294

- All classroom observations and reported estimates are synthetic.
- Slides use the current Unit 3 notebook’s data and calculations.
- The source project includes vector figures, source data and an R script to regenerate them.

Use the slide–notebook map in `README.md` to move between the lecture and each notebook section.